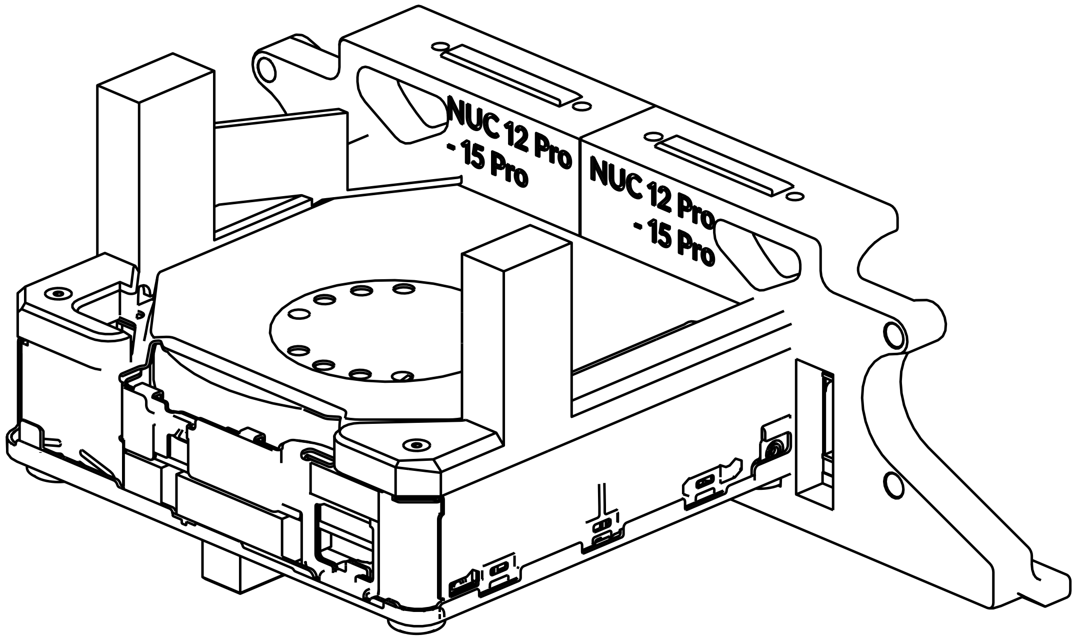
OitswilliamV2

ASUS / Intel NUC installation manual

REV 2026-02-04



OITSWILLIAM PANG

Hardware

Contents

Before getting started

You are about to read OitswilliamV2 NUC installation manual.

This mod attaches ASUS / Intel NUC computer into the printer in absolute back side of the Voron printer. Currently it only supports V2.4 150mm mod spec and have no intents of extending the support to other Voron printers, but will have intent of extending it to Micron Plus.

Guide and legend

Material - highlighted material / part

Material - mentioned material / part

Model model-variant-1.stl / model-variant-2.stl / model.step - printable / moldable model (these includes STL, 3MF or sometimes even source files like STEP)
The model name may not be mentioned in later pages.

Electronic parts will show their own power throughout this assembly guide, but here are the abbreviations that will be used:
V - volt,
A *or I* - amp (I as amp may be used but not in this manual.),
GND - ground.
In DC,
+ - positive,
- - negative
In AC,
L - live
N - neutral
⏚, E or G - earth ground
Other abbreviations include:
MCU - microcontroller unit
SBC - single-board computer

Warning

Danger: unless if it only runs on DC and only has a DC input, otherwise risk of electric shock

- Even though you can build with this mod from start, if you want to wire, or re-wire, you need to make sure and check that the mains is unplugged.
- Failure doing so which is unplugging the mains power may cause electrocution or even death.
- When in doubt, please don't touch the wires connected to the power supply or SSR.

Caution: static electricity. It is recommended to wear with at least the compliant wrist strap (may be it with 1-megaohm resistor from contact to earth) that is properly connected to earth.

Part printing

The parts of this model can be printed in the base color of the printer. **Bold text** indicates the recommended value and *italic text* indicates on what setting / value it needs to set.

FDM / FFF or SLS	Print material: ABS, ASA or PA12, **prohibits PLA.**
Layer height of **0.2mm,** initial layer height can be **0.2mm or higher.**	*Nozzle and extrusion width* must be **forced 0.4mm; if using 0.4mm nozzle**
1.6mm *wall width*, or **4** *wall counts.*	*Infill* must be **at least 40%, except on the infill modifiers**.
At least 4 *solid bottom and top layers*, **except on the infill modifiers**.	The best *print speed* is up to **120mm/s**. The speed might be reduced by the hot end flow rate.
Supports are **recommended for main skirt parts.** Use **tree or organic supports** if available.	*Other settings might be depended on your printer.* *Follow it's warnings for proper handling.*

Additional maker tools

To get started building the project, you may use the following tools:

- Wire crimping tool
- Soldering iron
- Heat shrink tubings, in 5mm diameter, one color minimum, two colors recommended.

Recommended Intel mobile CPUs (and AMD Ryzen mobile CPUs) for tasks

Task	Celeron (4c)	Core 3 (2p4e)	Core Ultra 5 / Ryzen 5	Core Ultra 7 / Ryzen 7
Run Linux with desktop environment	Suitable	Suitable	Suitable	Suitable
Can I run Windows natively and Klipper virtualized?	No	No	Probably, but we don't want planned obsolescence to ever exist, so no	Yes
Stronger than average computers, such as schools'?	No	No	Yes	Yes

To those who will be saying "I should get one with Celeron CPU," the problem is that Celeron often is trash, well only if we give Windows native to it. Beside this, Core 3 might have the same issue as Celeron's so it is not recommended to run Windows natively and Klipper virtualized.
CPU IDs: Celeron (4c): N5105, Core 3 (2p4e): 100U (this doesn't have low-power efficiency cores, let alone ray-tracing enabled GPU), Core Ultra 5: 125H (4p8e2l), Core Ultra 7: 155H (6p8e2l). Core identifiers p=performance core, e=efficiency cores, l=low-power efficiency cores, c=generic cores.

Recommended parts

This mount is intended to install NUC 14 Pro (slim version) or later, although older versions that is using 12th gen Intel Core or later still work. Other versions or variants may not fit properly, like need reverting for Pro tall version / Pro Plus, or just incompatible with Pro AI or ROG NUC.

Compatible	Incompatible
NUC 12 Pro, 13 Pro, 14 Pro, 15 Pro, slim version (models with K)	NUC 12 Pro, 13 Pro, 14 Pro, 15 Pro or so on, tall version (models with H)
	NUC 14 Pro Plus, 15 Pro Plus and so on
	NUC 14 Pro AI
	NUC Essential, doesn't matter whether it's Intel or ASUS
	NUC Performance (ASUS), ROG NUC and NUC Enthusiast (pre-ASUS)
	NUC Pro X / NUC Extreme
	NUC 11 Pro or earlier generations (normally, please request to us or create from blank.)

Note that this selection is very picky, if there is a variant selection for NUC Pro, choose the **slim** version.

Incompatible NUC models

- **NUC 11 Pro**
  - Did not have second method of powering. See connectors used.
    - This has already required modifications for the skirt.
- **NUC 12 Pro, 13 Pro, 14 Pro, 15 Pro, tall version (models with H)**
  - Device too tall, needs reverting to slim version
- **NUC 14 Pro Plus and 15 Pro Plus**
  - Form factor too big, needs reverting to Pro slim version
- **NUC 14 Pro AI**
  - Device seems only be intended for personal computing on the desk.
  - Different form factor.
  - Loses front panel connection.
  - Device doesn't seem to be intended for embedded solutions.
- **NUC Essential, doesn't matter whether it's Intel or ASUS**
  - Different form factor.
  - For NUC 14 Essential or later, it loses front panel connection.
- **NUC Performance (ASUS), ROG NUC and NUC Enthusiast (pre-ASUS)**
  - Different form factor and too big.
- **NUC Pro X / NUC Extreme**
  - Device seems only be intended for personal computing on the desk.
  - Different form factor, the computer assembly is card-based. The compute unit is too.
  - Device doesn't seem to be intended for embedded solutions.

- **For NUC Performance (Intel), since they have different port arrangement and we weren't intended to make that, so we have to put the lineup incompatible. And for NUC 11 Pro, it only powers via barrel jack, but we normally do not support it, so we have to put that lineup incompatible as well. If you need it, you can request for such compatibility or modify the blank mounting panel.**
  - **NUC Performance was the old name for NUC Pro, before the one with 11th generation Intel Core.**
- **For NUC 16 Pro, the form factor is different, and is on the works. The form factor requires changing the Z axes motors with Clockwork (first generation) motors first.**
- **For other (non-ASUS or Intel) "NUC"s, since they did not provide CAD references, we did not make for these.**
  - **For requirements:**
    - **Height limit of 36mm, excluding feet, or total of 40mm,**
    - **Width and depth limit of 130mm, if slotting to the skirt, or 150mm if changing motors be the requirement.**
    - **Must support "backpack" via VESA monitor mount.**
    - **Unless if the NUC can't slot into the skirt, the NUC must slot in 8.4 to 8.2mm to the skirt (or the IO shield must have at least 1.2 to 1.4mm of thickness).**
    - **If the NUC is not enclosed in exoskeleton (dual-casing design), it may be difficult to be placed into the skirt, and may rely on legacy back skirt.**

Connectors used

- DC2 (Molex Micro-Fit 3.0 2*2P connector)
  - Used to power the NUC if not using with the included barrel jack power supply.
- Front panel power button (DuPont 2.0mm pitch, 2-pin recommended, connector that utilize all pins will render the bottom cover unusable)
  - Used to power the NUC on via RealEstate display.

Other information

- You may read the service manual for your NUC through ASUS Support or the vendor that make such mini PC. This manual may not serve some information or material where the service manual has written.
- RealEstate G2.1 from OitswilliamV2 mod is recommended prior installation. This unit comes with the power button that can be connected to the PC / SBC.

About OitswilliamV2 NUC mount

The NUC mount is one of the key parts of OitswilliamV2. This is one of the stupidly insane task as no one has thoughts about, yet magical, it grants printers performances no budget laptops can achieve. It uses ASUS NUC Pro as it's base and mounts the computer to the back of the skirt. It will power the future of user experience.

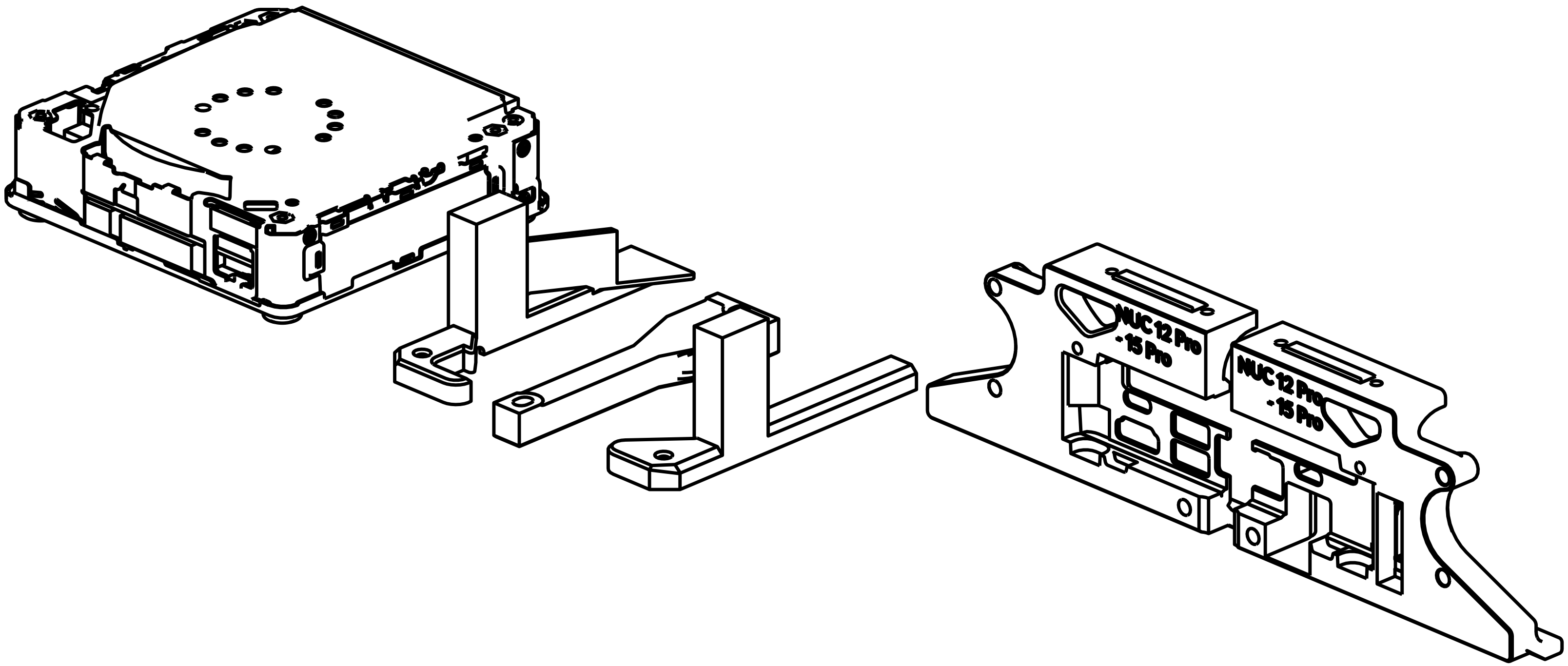
Contribution

This mod is a part of OitswilliamV2 mod project through GitHub. You can visit, contribute and report new issues here.

Technical support

You can contact Oitswilliam here, or if the issue is not related to OitswilliamV2 mod, you may visit Voron forum or its Discord server.

Assembly



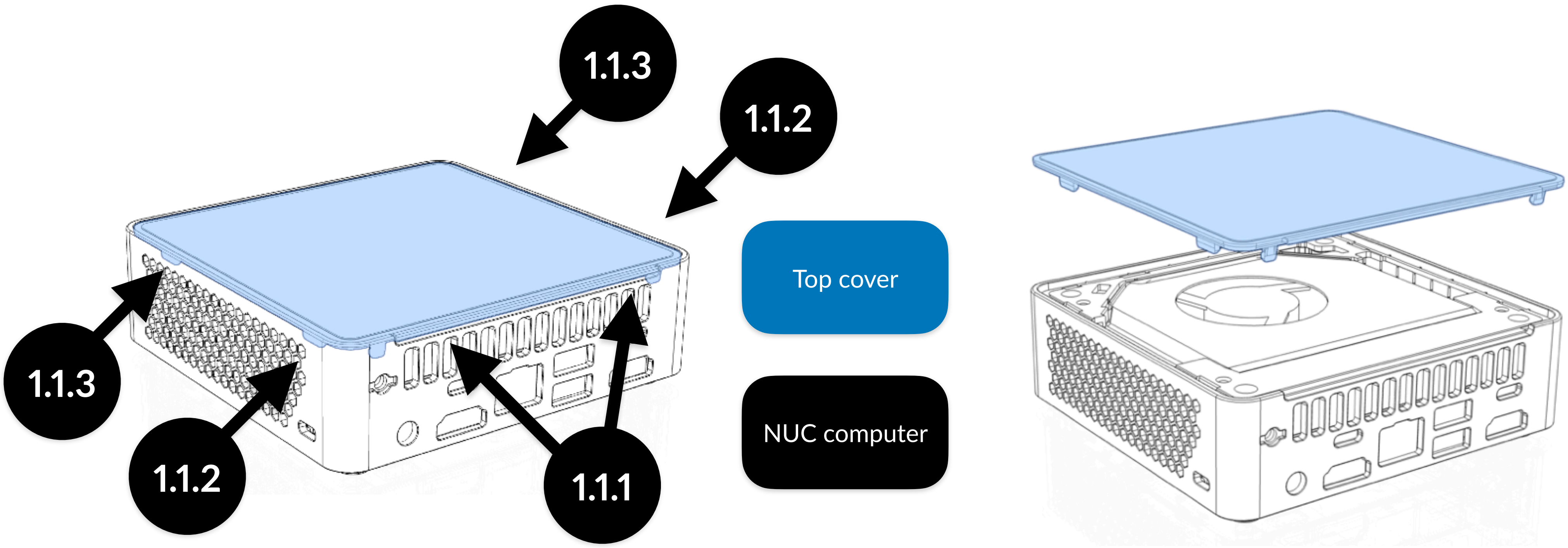
Modify NUC

The NUC needs to be modified before being attached to the skirt and be put on the printer. If you do not do this procedure and just put it on the printer, then it is too late and needs disassembly.

Make NUC top-naked

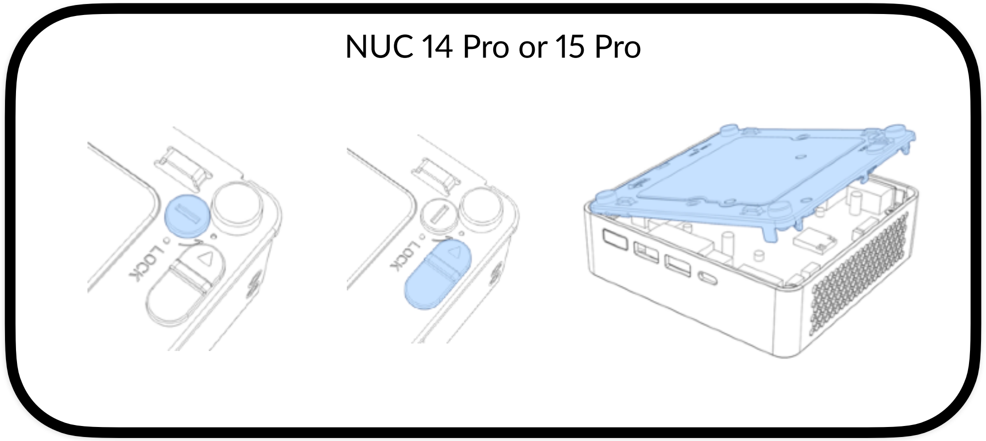
The design of the NUC mount requires the PC to be top-naked, to improve its cooling when housed inside the printer.

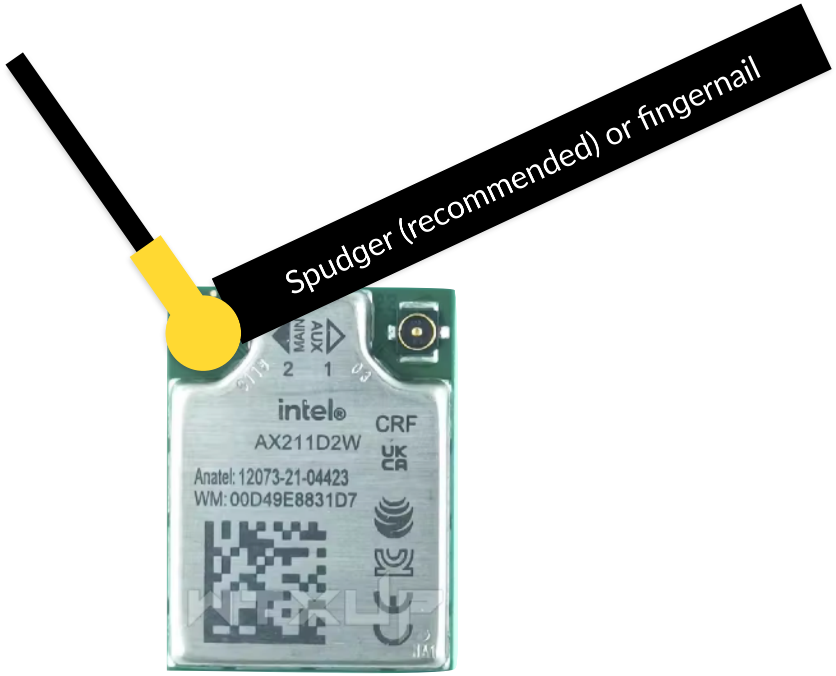
1. Unclip the top cover



To unclip the top cover, start prying at left- and right-back side between the top cover and the plastic case (1.1.1), then go from back (1.1.2) to front (1.1.3). When unclipped correctly, the top cover can be removed.

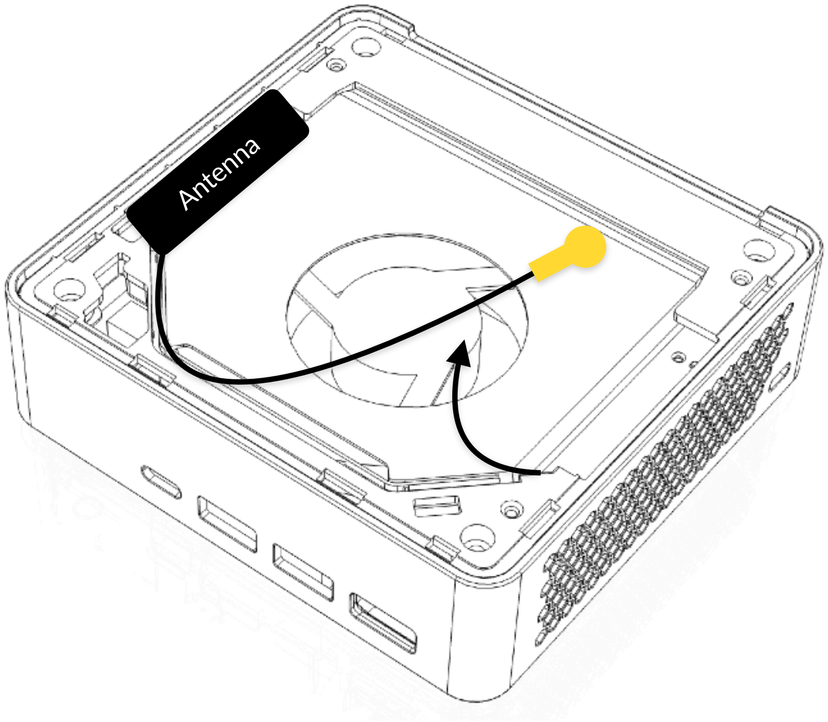
2. Disconnect the antenna cable (ASUS NUC 14 Pro and NUC 15 Pro)
If you are using NUC 14 Pro or NUC 15 Pro, you need to disconnect the antenna with the black cable (or as technicians call it main antenna), otherwise you will break the cable removing the plastic case. For Intel / ASUS NUC 13 Pro or earlier, this is not required.



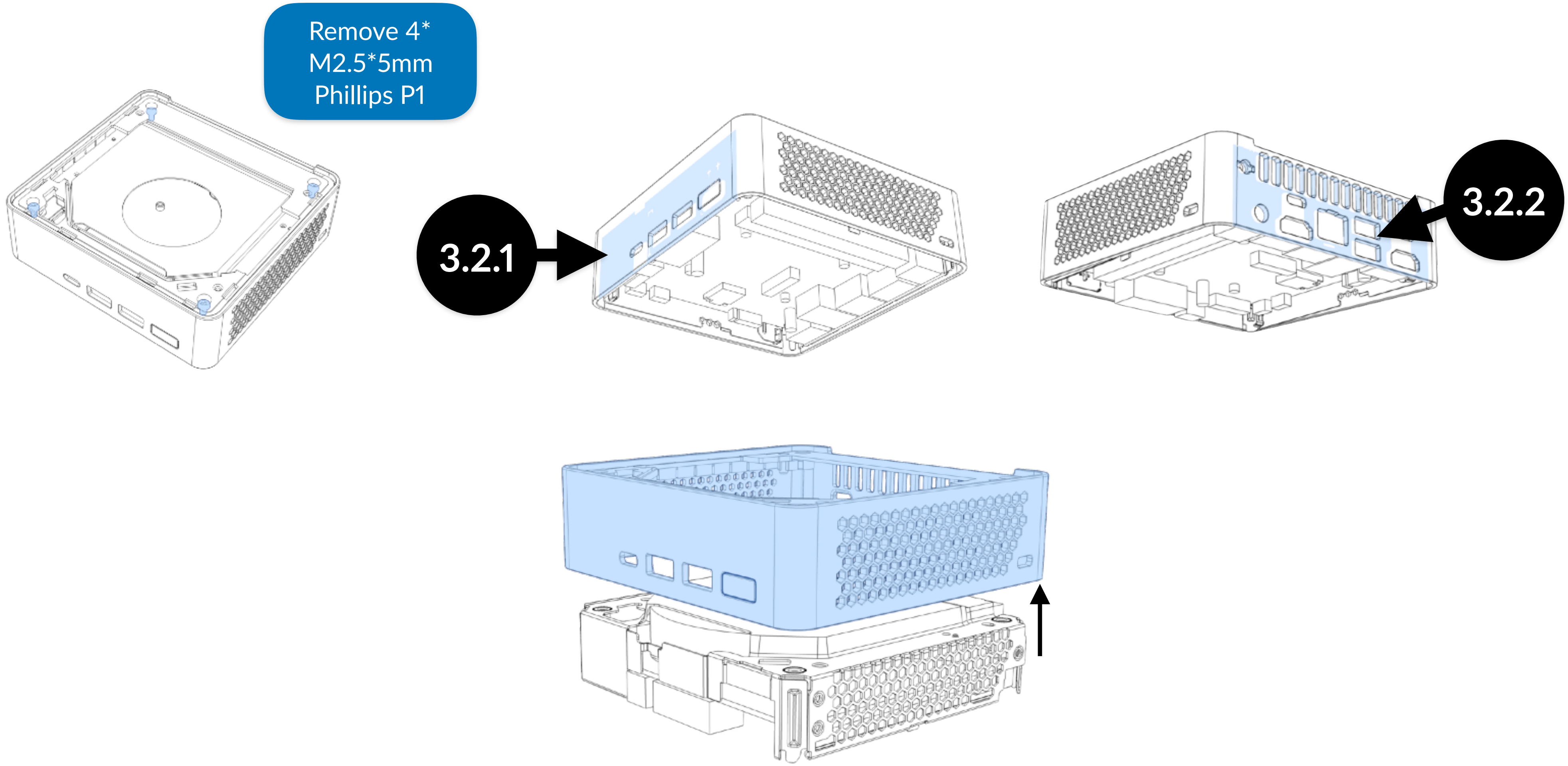


This step is not required if you have ASUS / Intel NUC 13 Pro or earlier~~, or ASUS NUC 16 Pro~~.
Open the bottom cover, unlock and push the latch, then disconnect the black cable from the Wi-Fi + Bluetooth transceiver with the Spudger (recommended) or finger nail.

After that, get the black antenna cable out of the shell. Carefully pull the cable up from the plastic housing.

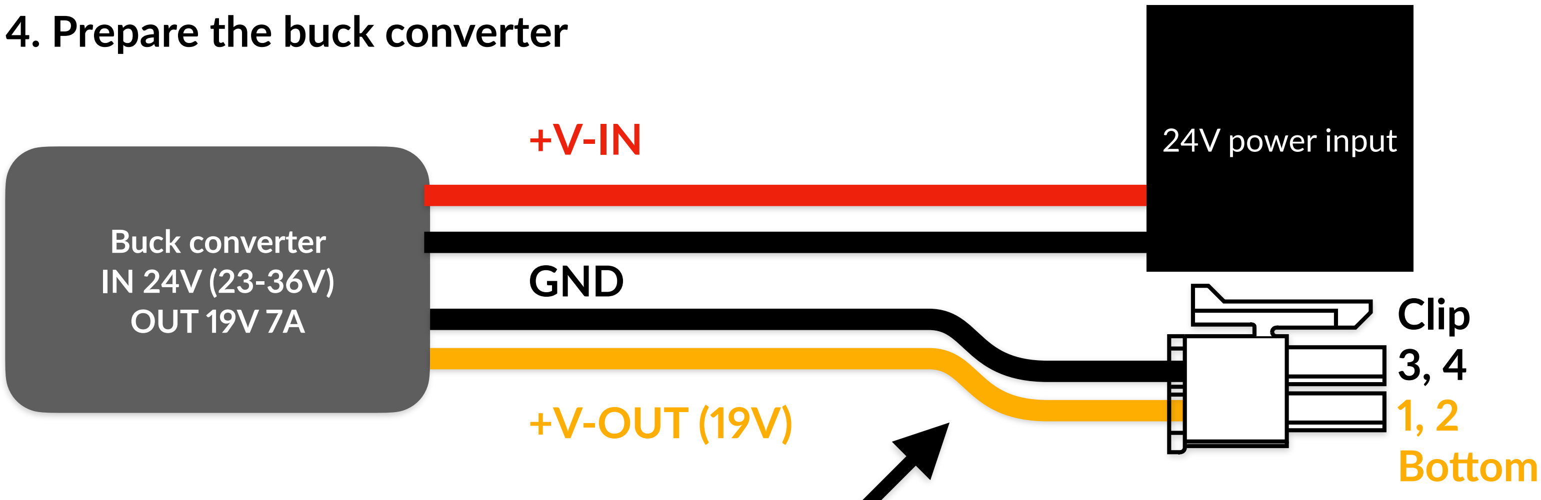


3. Remove the plastic housing

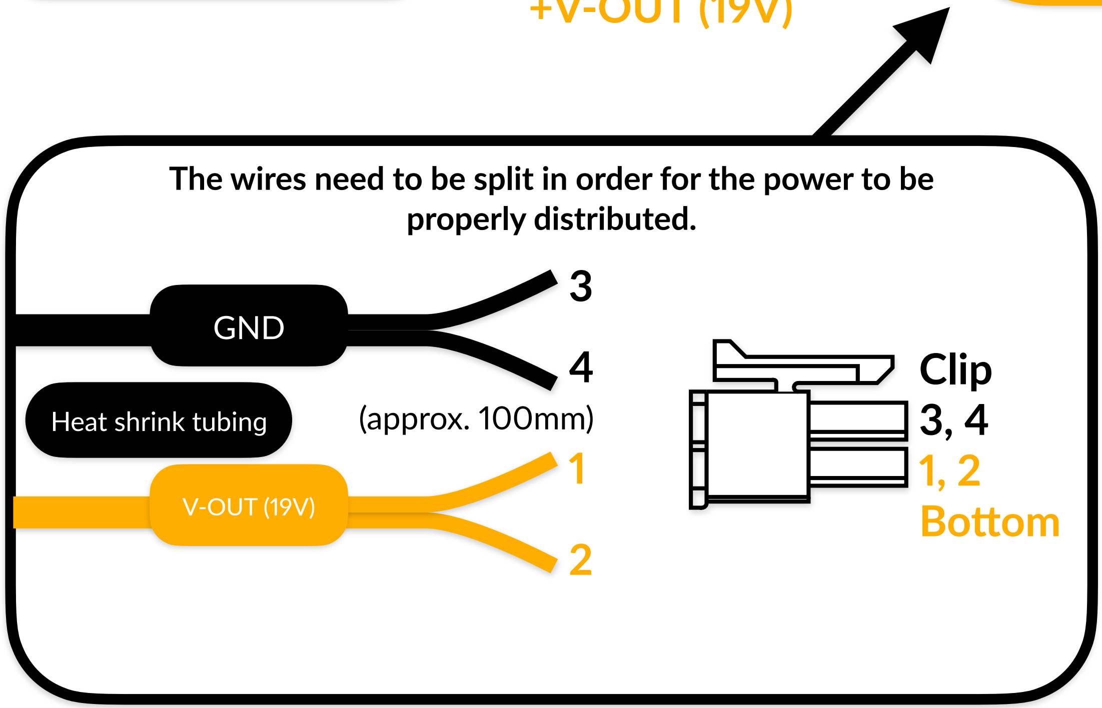


Start by removing four M2.5*5mm Phillips P1 screws, then pitch and unclip the front side first (3.2.1), and then unclip the back side (3.2.2).

4. Prepare the buck converter

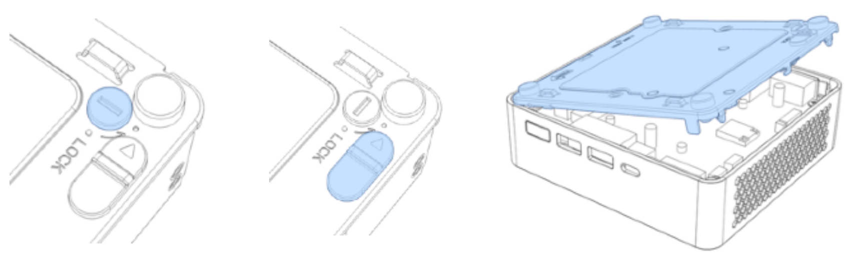


The NUC can run in voltages from 12V to 20V, but just to be safe, a 19V buck converter is recommended.
The NUC variants may require different minimum current to function correctly.

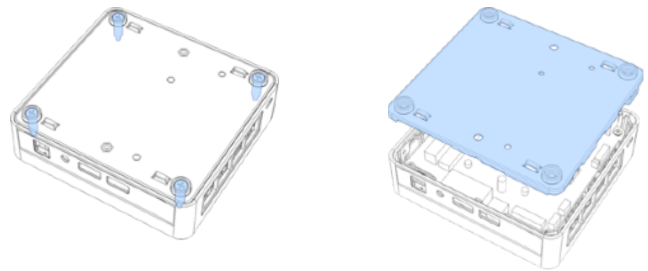


5. Remove the bottom cover, if re-attached prior

NUC 14 Pro or 15 Pro



NUC 13 Pro or earlier

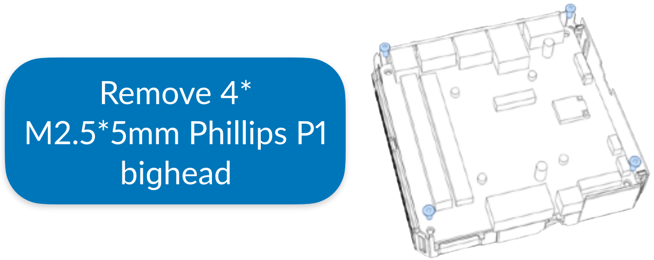


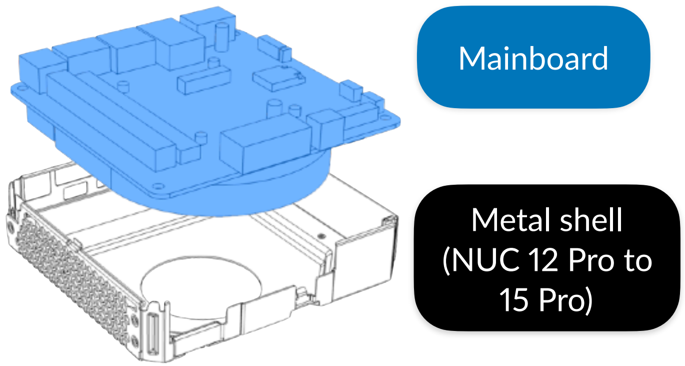
For NUC 14 Pro and 15 Pro, unless if the cover was stayed removed, push the latch and then turn the bottom to remove it.
For NUC 13 Pro or earlier, unscrew four 1315 (M2.5*18 captive) screws in legs, then pull the whole bottom cover to remove it.

6. Disconnect the antenna cable, main (NUC 12 and 13 Pro) and auxiliary cable(s)

Disconnect the white cable from the Wi-Fi + Bluetooth transceiver using the Spudger (recommended) or finger nail. If the black cable is not disconnected yet, it should be disconnected as well. This is to prevent damage.

7. Extract the mainboard (NUC 12 Pro to 15 Pro)

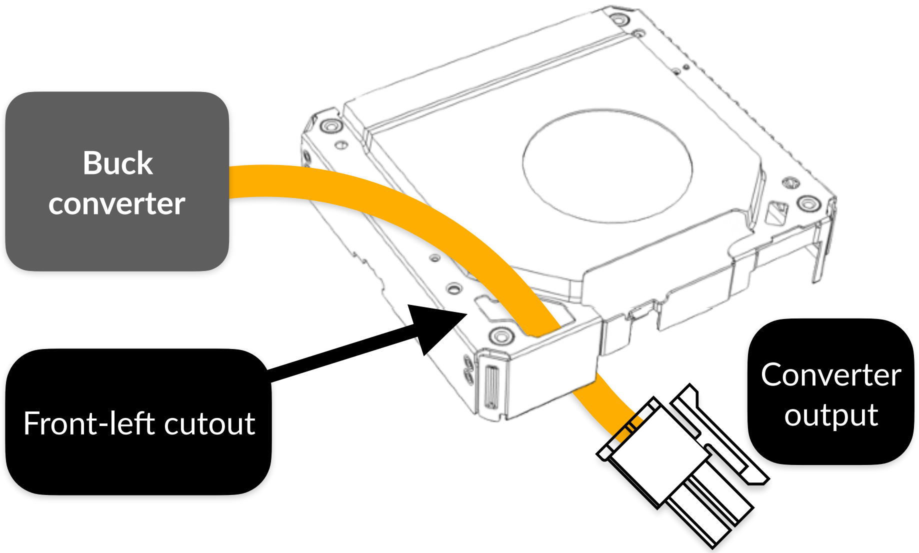


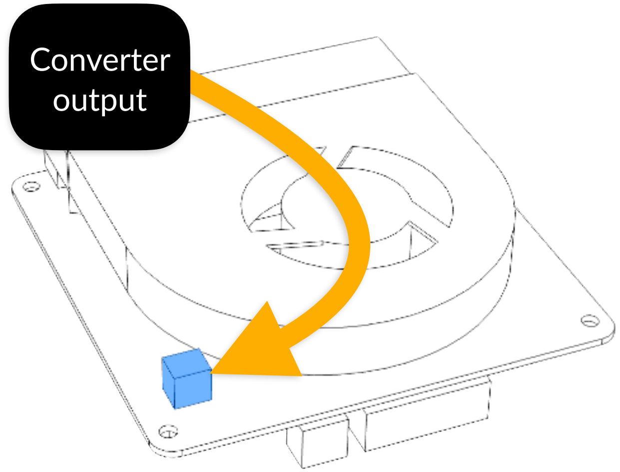


Remove four black M2.5*5mm Phillips P1 bighead screws that's on corners attaching the mainboard.

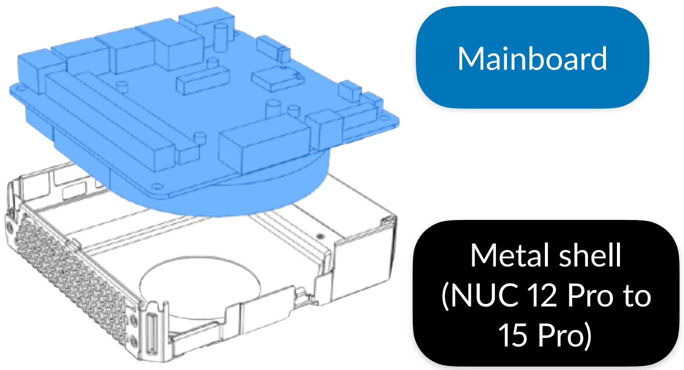
8. Buck converter to DC2 connection

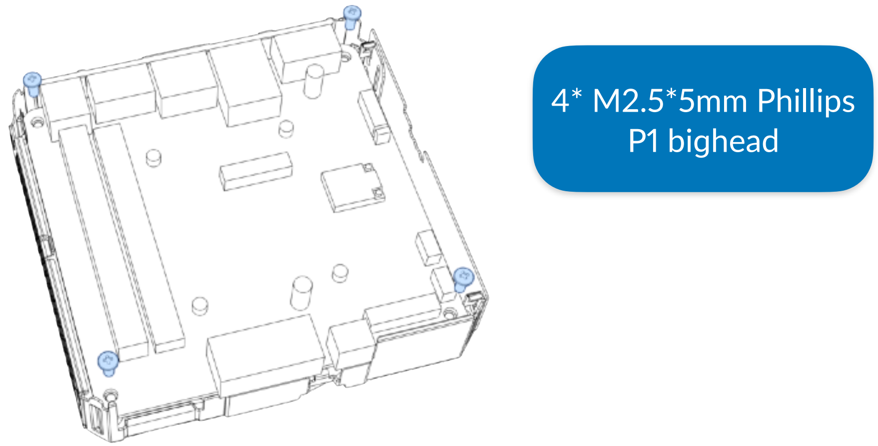
The power output cable needs to be passed through the metal shell before it is connected to the mainboard, otherwise the mainboard will be impossible to reattach. There is a cutout at the front-left side of the shell where the power cable can pass through. Once passed through, the power output can then be connected to the mainboard.





9. Reassembly, without the plastic housing, but with front panel power button extension

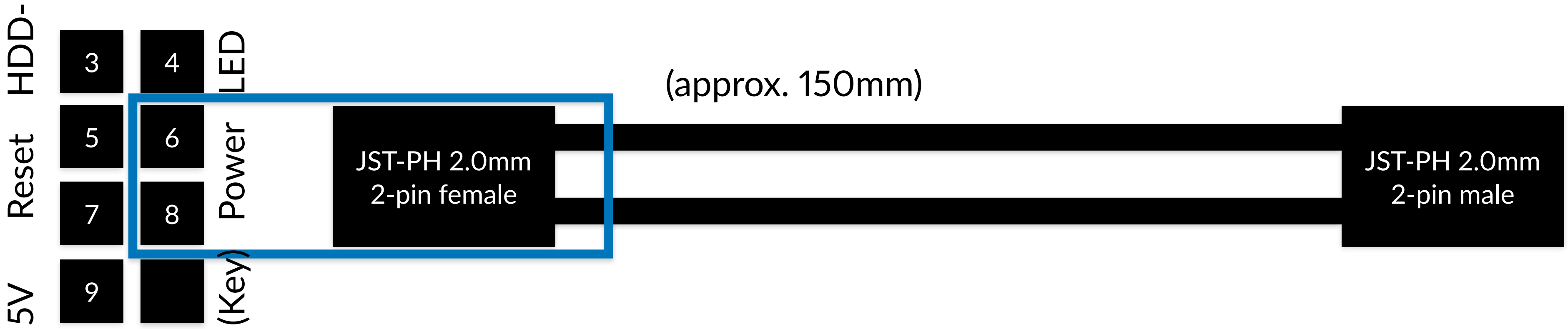




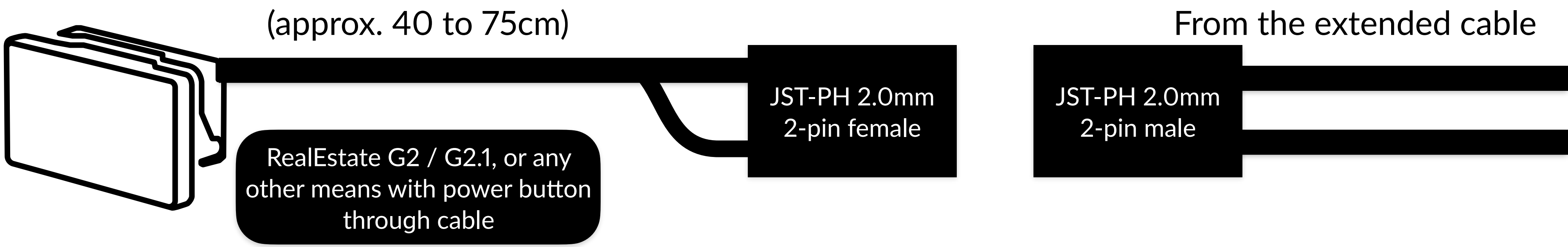
You may do the reverse of what you have opened, but without the front plastic housing. First, carefully reattach the mainboard (as the converter output is connected), and secure it with four black M2.5*5mm Phillips bighead screws,

Then, re-connect the antenna cables, Spudger may help, make sure the cable colors and the arrows match.

Create the extension cable



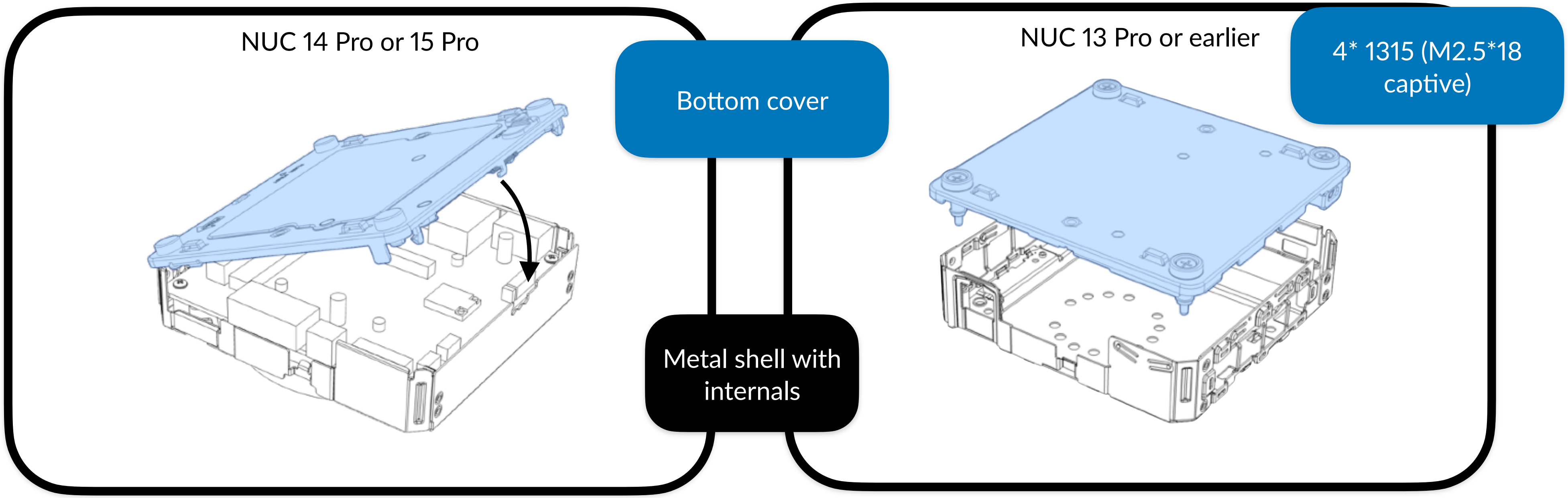
The polarity doesn't matter as we are just connecting to the button. You may connect the extension cable to the pins 6 and 8 of the front panel header. **The first pin will have an arrow pointing at, and the tenth will not have a metal pin.** The cable can be sled through between USB-C port and a USB-A port.



You may connect the power button cable to the front IO header extended by the extension cable after mounting NUC to the frame.

- And then, the M.2 PCIe NVMe SSD and DDR4 (NUC 12 Pro and 13 Pro) / DDR5 (NUC 14 Pro or later) RAM sticks can be loaded, if not loaded prior.

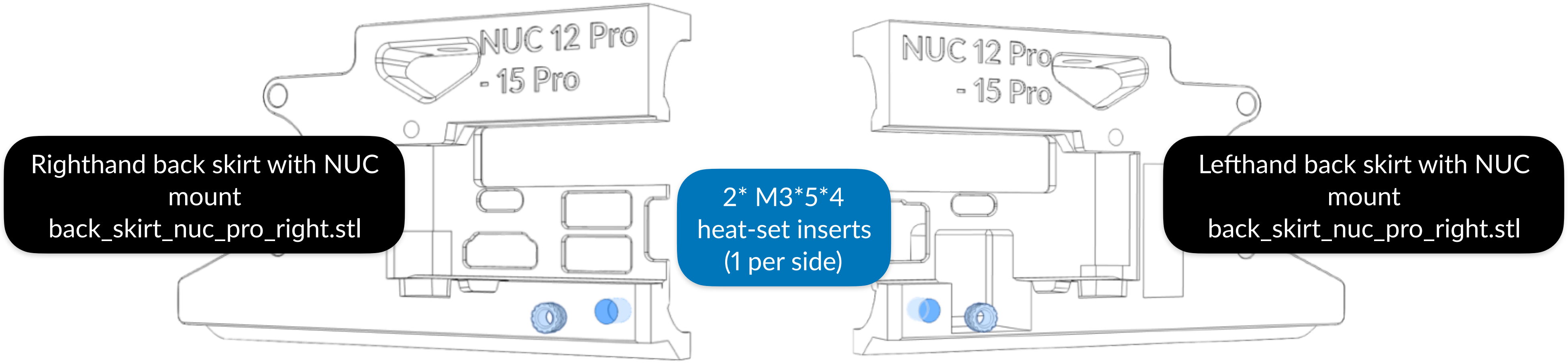
Since NUC has two USB 2.0 headers as in Molex PicoBlade headers, You can create up to two of your own Molex MicroBlade to USB 2 Type-A adapter using Molex MicroBlade pigtail cables, female USB 2.0 Type-A ports and a custom shell. Please refer to ASUS' or Intel's NUC, or the vendor's technical product specifications for the pin types.



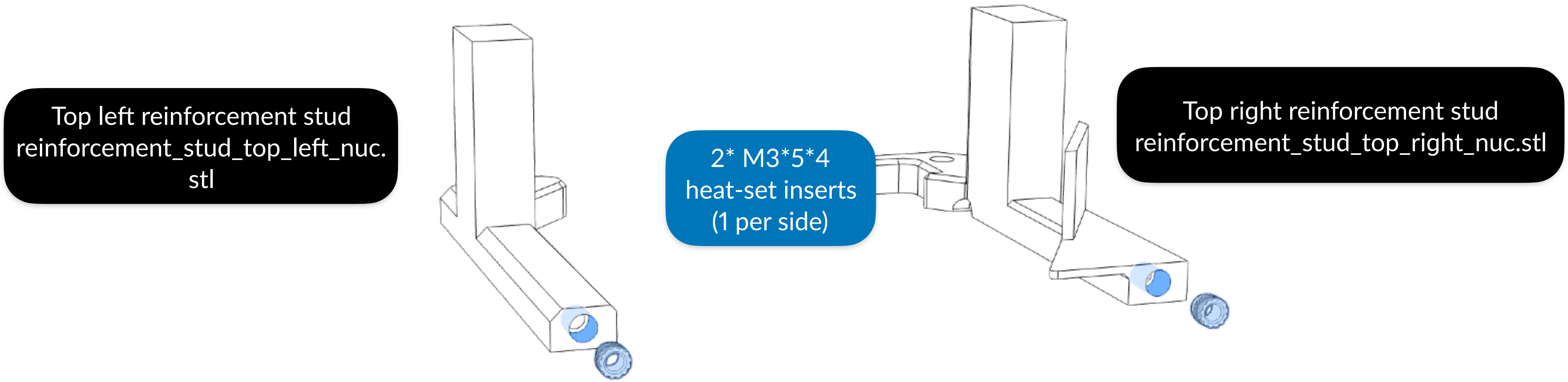
And finally, the bottom cover can be re-attached (and locked). If the bottom cover has thermal pads for the corresponding drives, you may let it use them.

Skirt preparation

There are additional heat-set inserts that will be placed at the bottom back face of the skirts, not just the front face and the bottom from the page 212 of Voron 2 assembly manual.

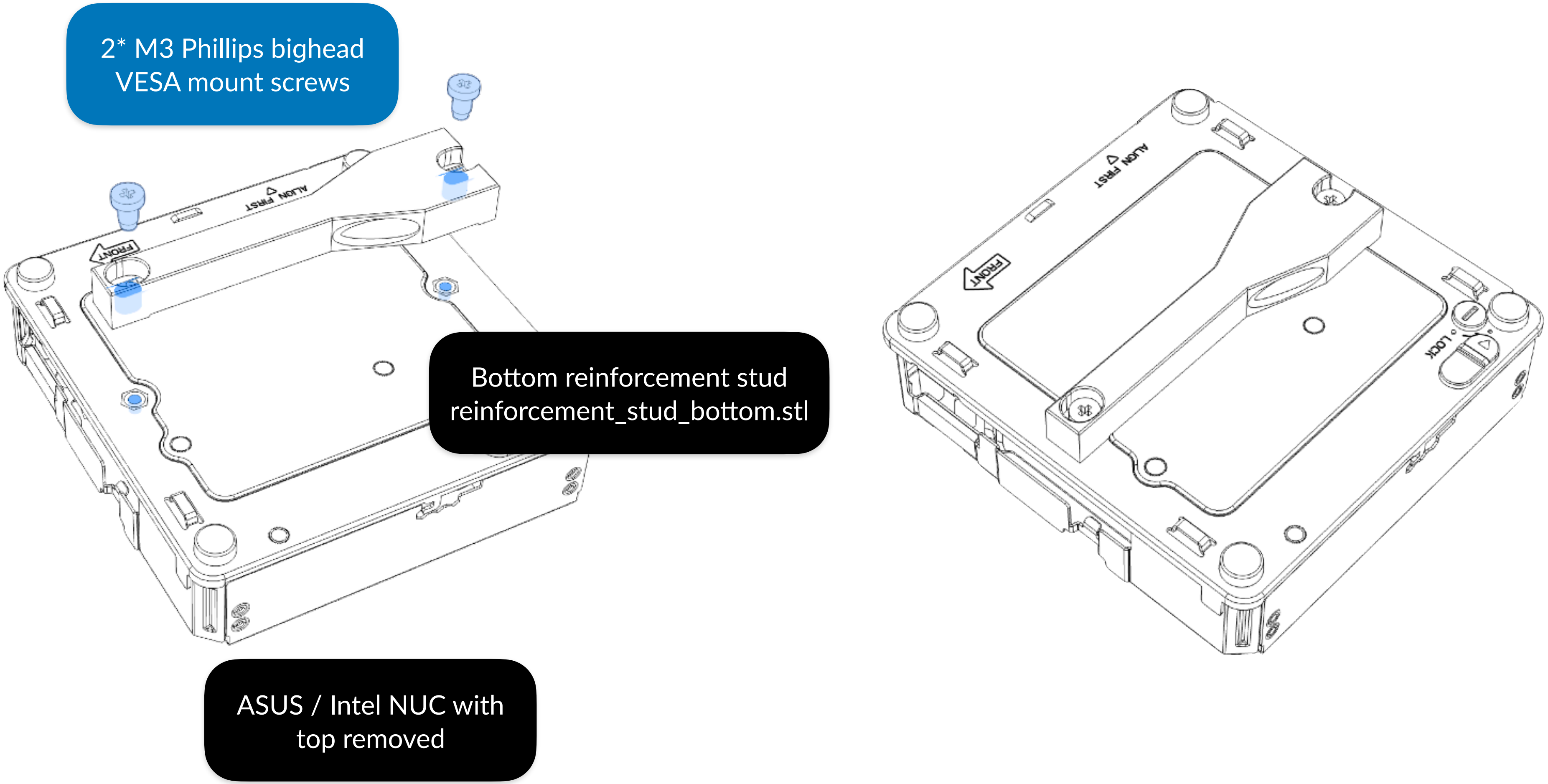


The top reinforcement studs also need heat-set inserts as well.

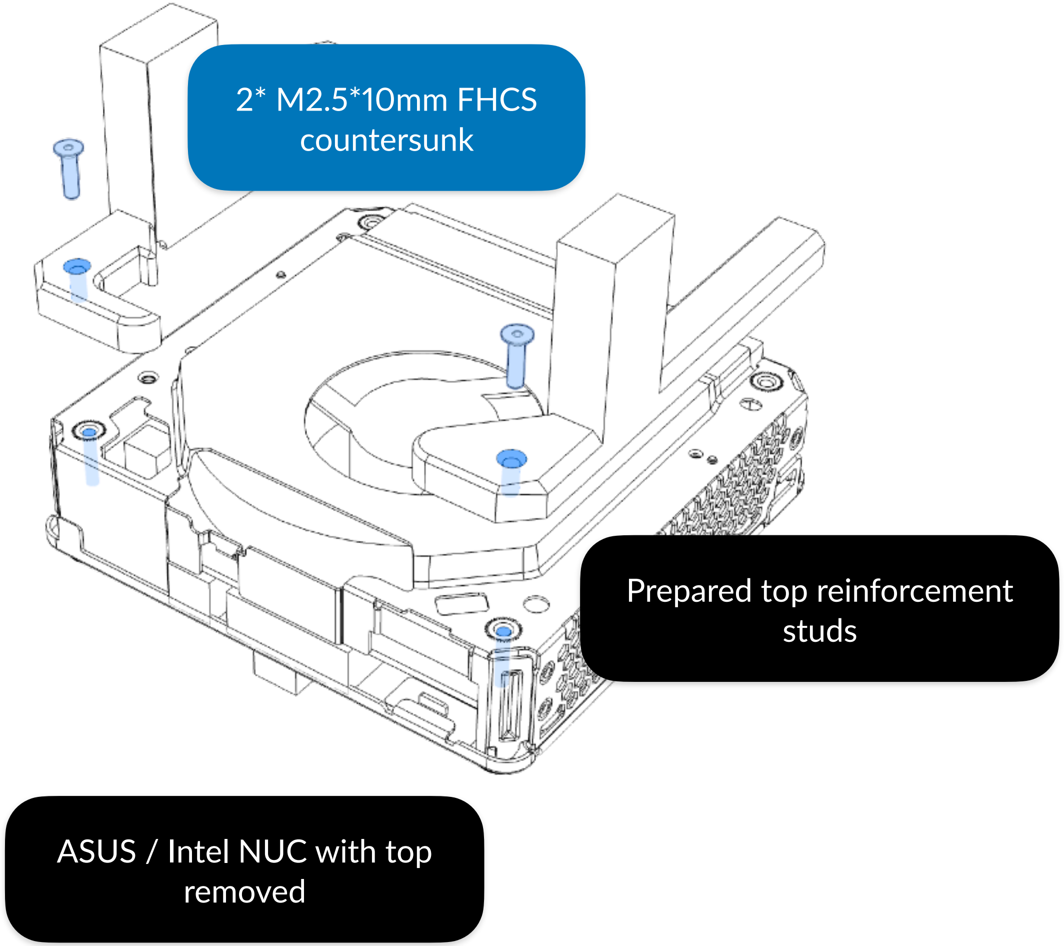


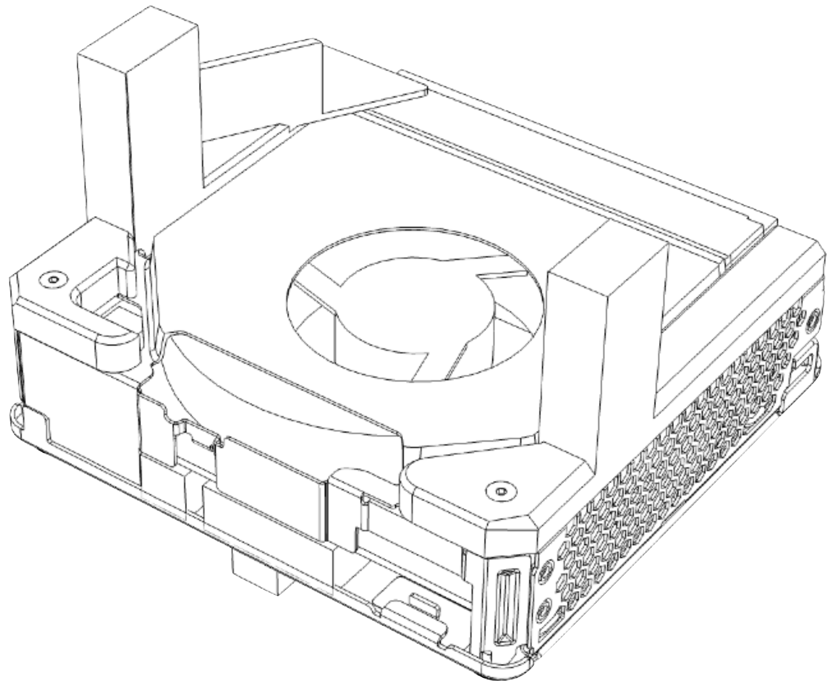
Install the module (ASUS / Intel NUC)

The bottom reinforcement stud requires two Phillips bighead screws intended on making NUC mountable on monitors. We will be using these as the mounting screws for the stud.



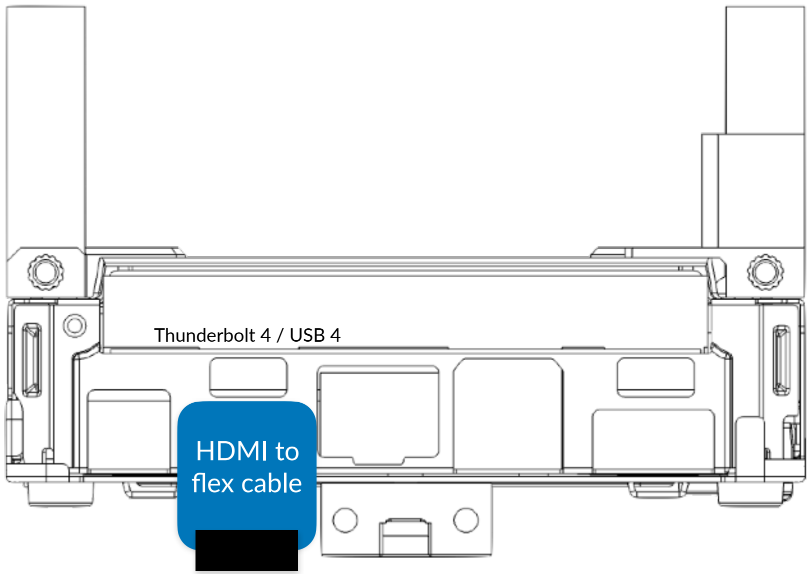
The top reinforcement studs, on the other hand, requires one M2.5*10mm FHCS countersunk screws on each side, totaling on 2. The left has a cutout for the wires which allows the procedure without complexity.

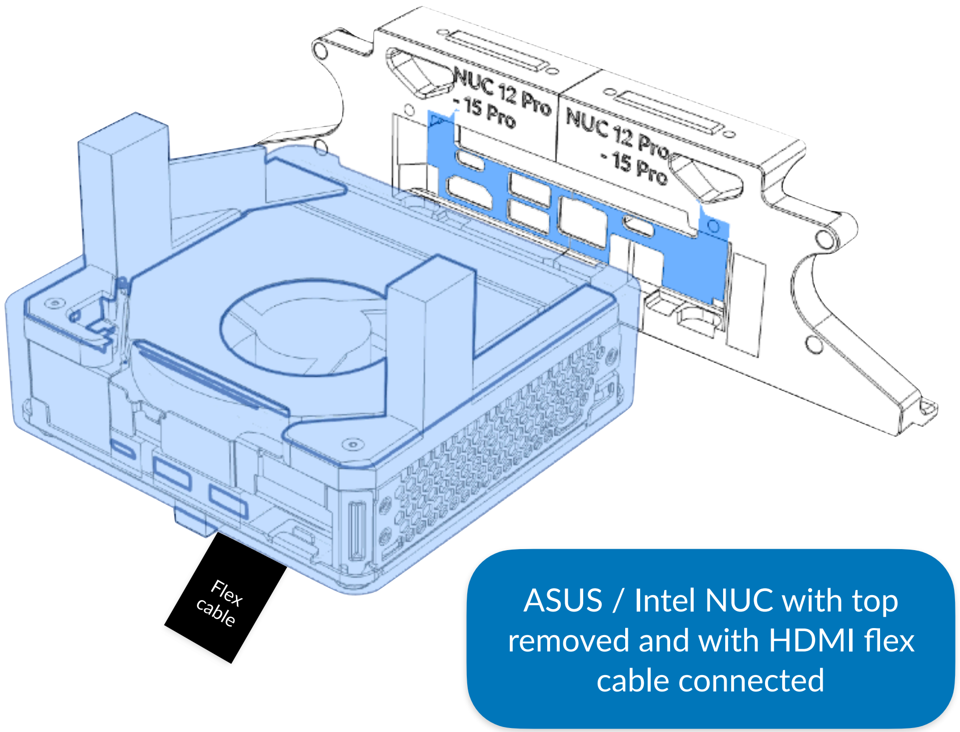




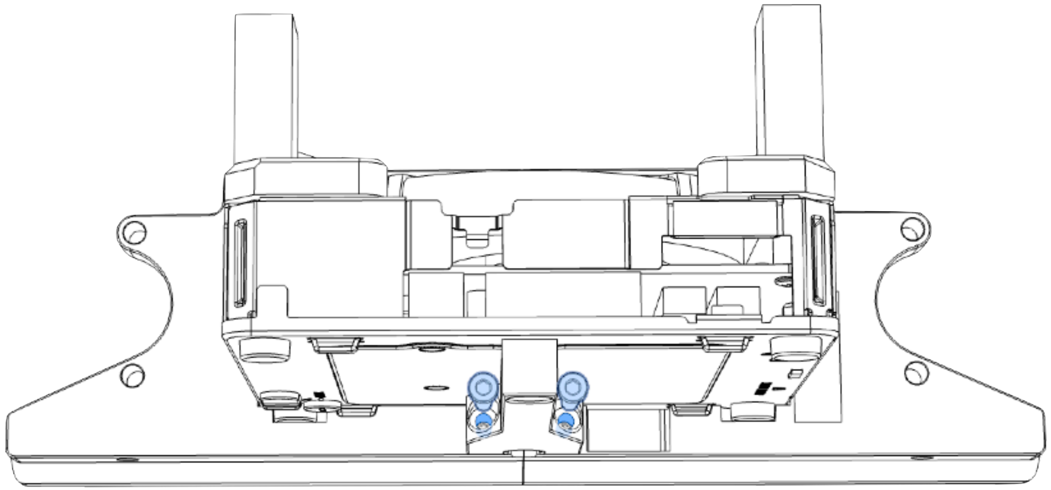
The HDMI to flex cable and the FPC cable needs to be connected to the left side of the HDMI port before it is being inserted to the skirt.

If the skirt is being attached for the first time, the NUC needs to be inserted first, and then attach the skirt to the frame. Otherwise the screws between the NUC and the skirts may not be properly aligned.

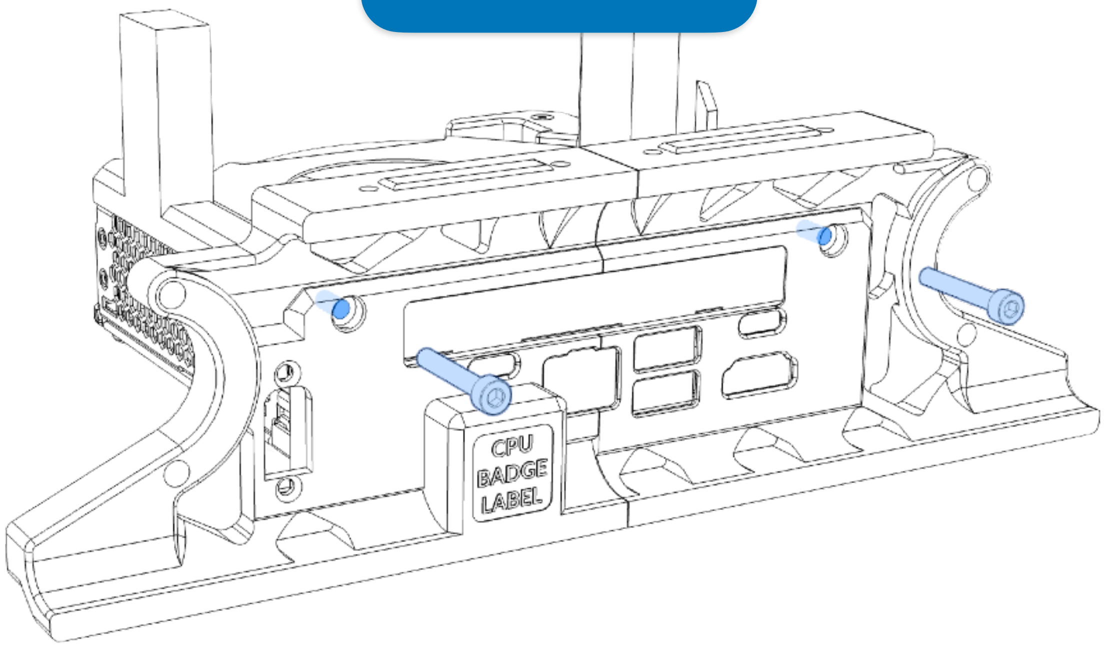




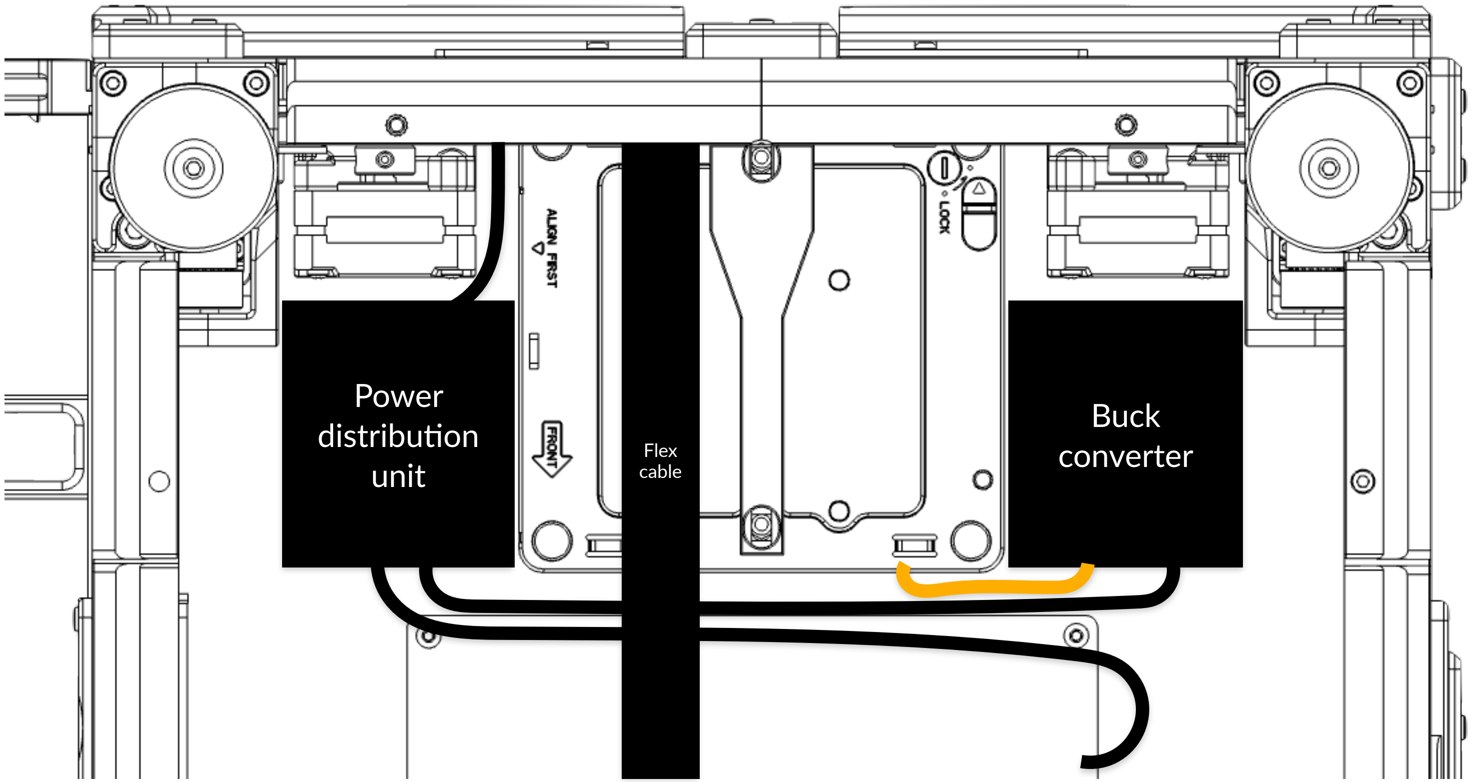
Securing NUC to the skirt



4* M3*20mm SHCS
(2 per side)

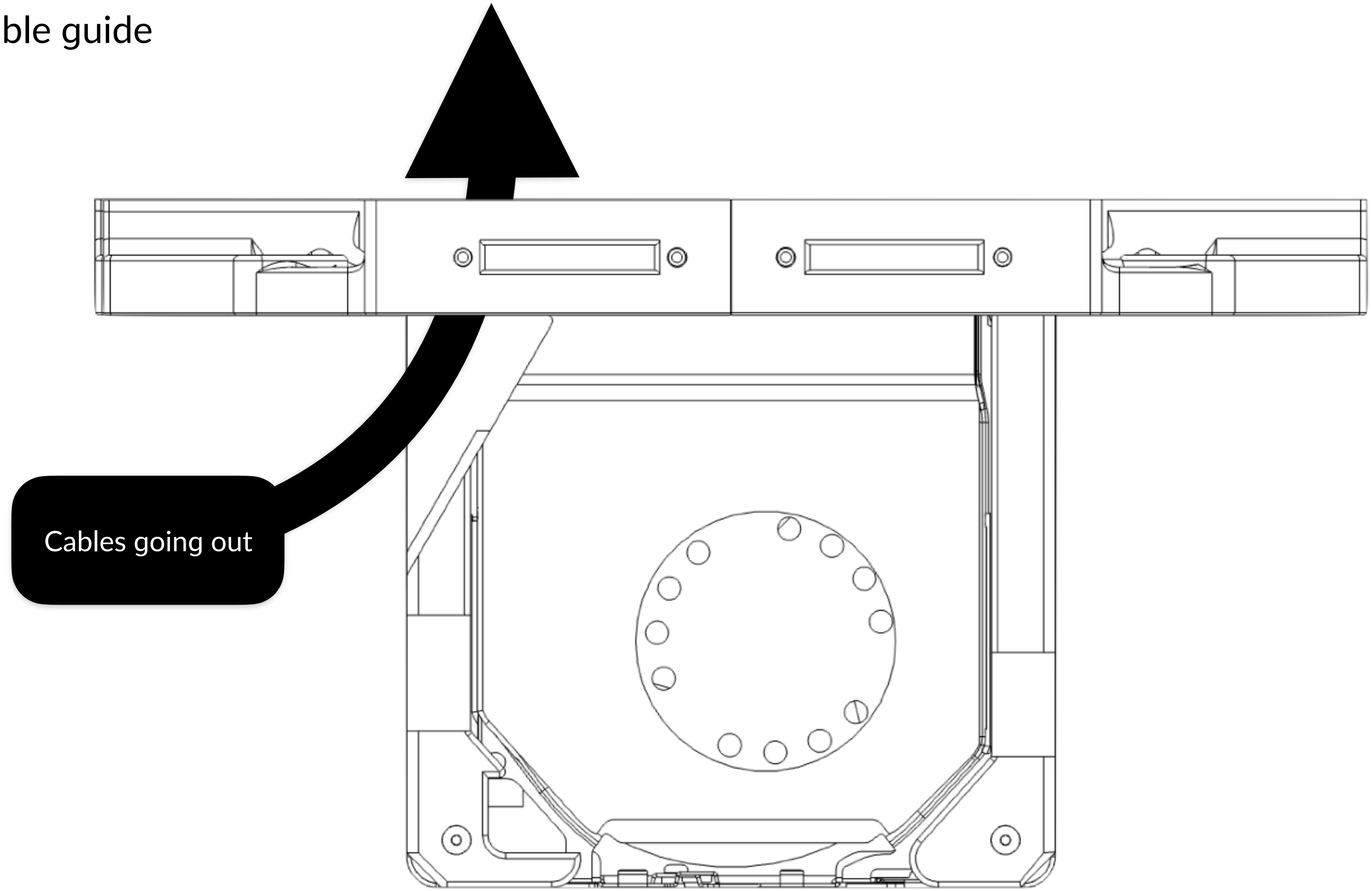


Mounting to frame
Organization guide

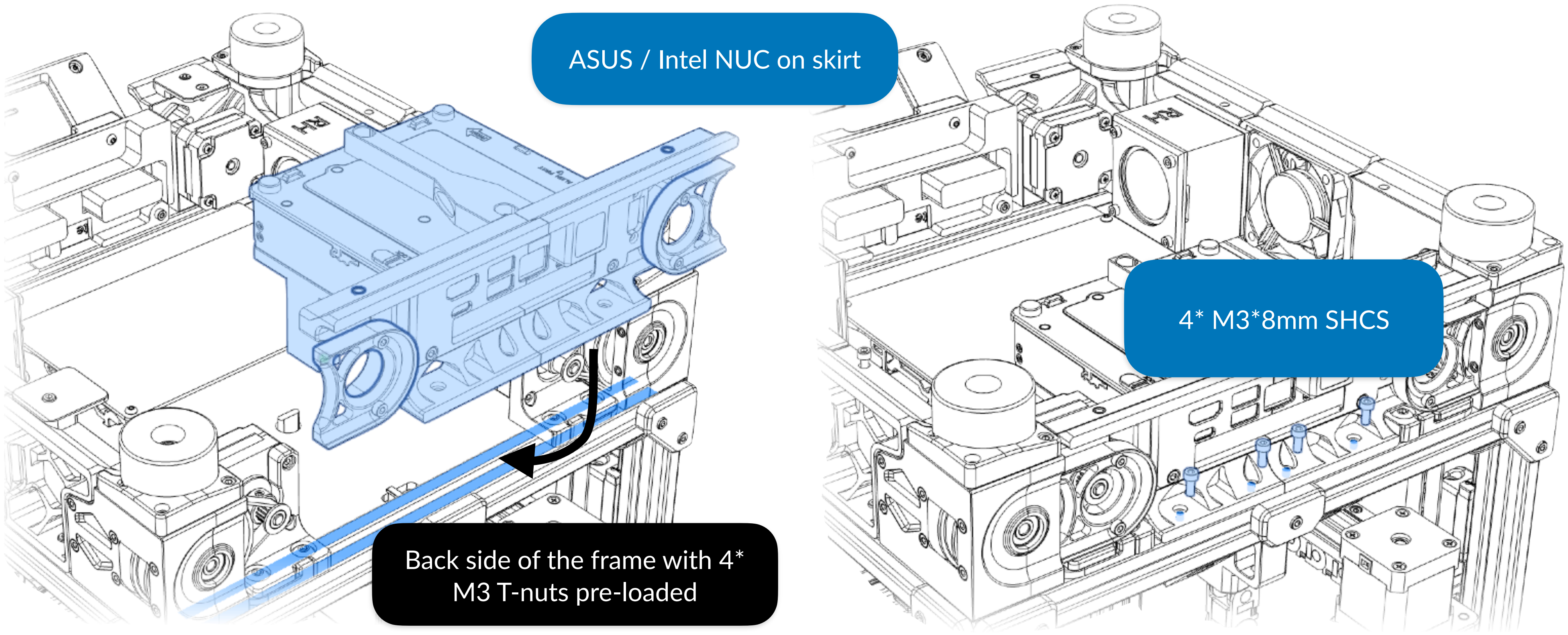


The power distribution unit can be placed at the left side of the printer, near the power input connector, while the buck converter can be placed at the right side of the printer.

The cable guide



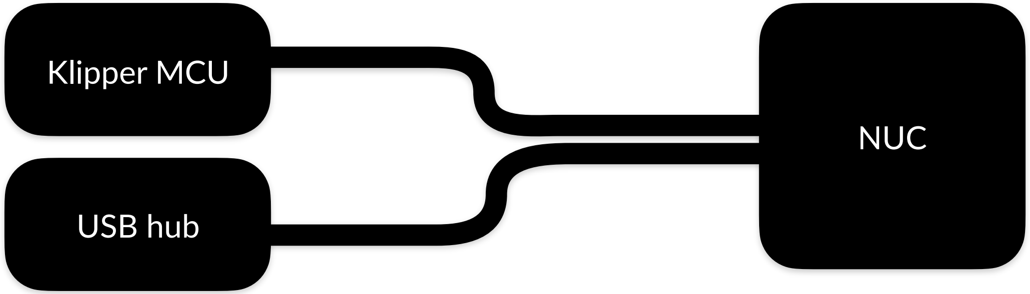
The top-left reinforcement stud has a guide that is made specifically for the cables going out of the skirt and which prevents the cables from covering the NUC's ventilation holes. Cables that are going out from the right side should be placed or re-routed to the left side.



To properly slot the NUC skirt to the frame, slot it where the unit is slightly out of the frame, and then slide it in. Then align the T-nuts and use four M3*8mm to secure the NUC skirt to the printer. Note that some screwdrivers may not properly fit as the spacing between the screw and the IO panel are close. Attaching XT60 power input requires the skirt to be attached to the frame first, and then such connector can be attached using two M2.5*10mm FHCS countersunk.

Hardware connection guide

You may connect non-Klipper devices to the USB 3.0 hub. Avoid connecting Klipper MCU or CAN bus unit to the USB hub that isn't intended for Klipper MCU as the hub may easily be overloaded.



You can then connect the power button from the front panel header that was extended prior. After that check to make sure the electrical work is done before covering the electronics down with the bottom panel.

Software configuration

Debian Linux physical (including Ubuntu)

You can install Debian Linux to NUC. Virtual machine is not required. It is recommended to install the latest version of Debian or Ubuntu with the desktop environment. After that, you may create own user account, setup APT server communication, choose the features. Once Debian or Ubuntu is installed, you may install KIAUH from dw-0 if you want connivence, install Klipper, lightweight UI and probably Crowsnest through KIAUH.

Windows 11 physical + Debian Linux VM

If you are a diehard, you can install Windows 11. However a virtual machine is required to run Klipper as it is Linux Python-exclusive software. This also means requiring a Debian ISO prior install on virtual machine.

This is not recommended, but anyway, we are diehards

This method is not recommended as Klipper heavily relies on real-time scheduling with up to 100 milliseconds of tolerance and virtualization tends to exceed that. However, there are workarounds, such as Process Lasso from Bitsum, which allows reserving CPU cores, put programs in priorities and limit programs to only use said threads with at least three changes.

Windows Subsystem for Linux (WSL)

If the NUC has Windows installed (and licensed), you can enable this feature and install WSL, install Debian / Ubuntu Linux and use KIAUH inside WSL.

VirtualBox

You can use VirtualBox if you don't want WSL, or wanted a more graphical way to adjust VM settings. However there are recommended settings that has adjusted, you need to shut the VM down in order to change:

- 1GB of RAM, 128MB of video memory,
- Dual- (recommended) or triple-core,
- 3D acceleration enabled,
- A bridged network adapter and a host-only adapter,
- Virtual NVMe instead of virtual SATA.

If you want to create a virtual machine from scratch:

- First, specify the name of the virtual machine, you may call it **klipper**.
- Then you select the ISO image (not the photo) of Debian Linux.
- And then manually specify the virtual hardware, with 1GB of RAM and dual-core.
- And specify the virtual hard disk and make sure it is somewhere around 32GB and 64GB.
- The installation steps for Debian can then be proceeded like it is in the physical machine, such as creating user accounts, choosing features, installing Klipper and its UI.

Assembly complete

And we are done! If you are satisfied with such product consider tagging @officialbunny350 on Facebook or @realBunny350 on X (formerly known as Twitter) and share such product with us on Oitswilliam Pang Discord Server.

If you have enquires, problems or issues related to building or using such product, please report it on GitHub.com/Bunny350/OITSWILLIAMV2/issues and we will get on it.

Change log

- 2026-02-04 - Initial release

Manual designed by Oitswilliam Pang.
One-person operated design.
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